QMI/QDCS : ZX-Calculus

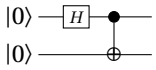
Renaud Vilmart

Exam

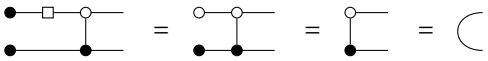
⚠ Disclaimer: unless specifically said otherwise, you can ignore the scalars and focus on the diagrams.

1 Simple circuits

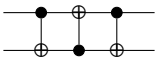
Question 1. Turn the following circuit into a ZX-diagram:



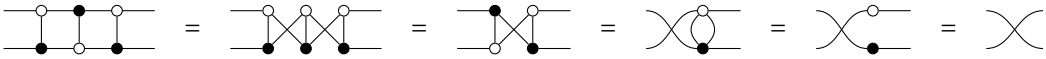
Simplify it to show it is equivalent to the cap (if reading the ZX-diagram from top to bottom). Provide the axioms' names used at each step.

Answer: 

Question 2. Turn the following circuit into a ZX-diagram:

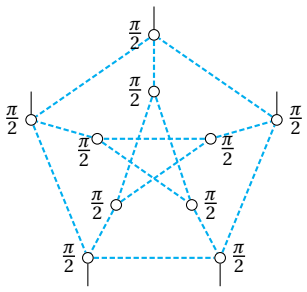


Simplify it to show it is equivalent to the swap. Provide the axioms' names used at each step.
Hint: You can either try to do it directly, or using a normal form followed by "obvious" simplifications.

Answer: 

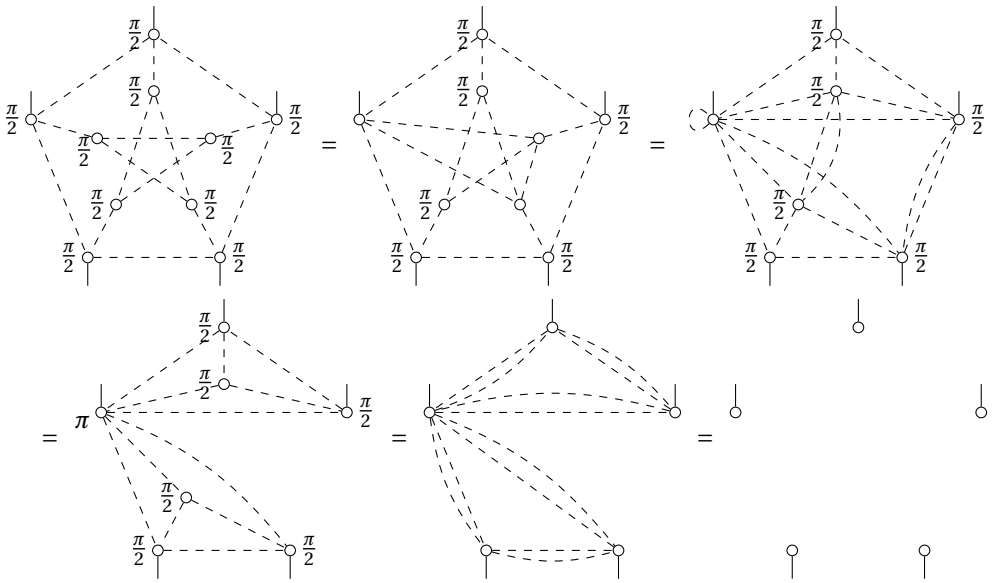
2 Clifford normal form

Question 1. Consider the following diagram in graph-like form (where the dashed edges are H-edges):



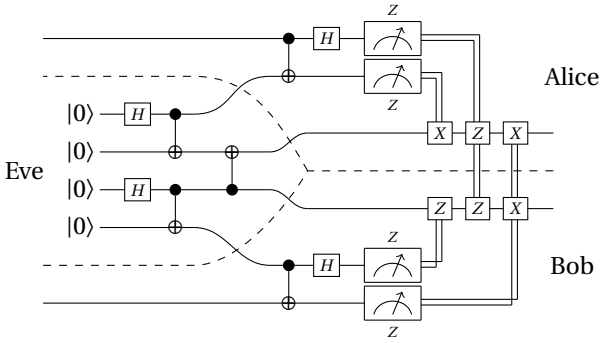
Put it in Clifford normal form: *Hint: be careful with potential common neighbours when applying a pivoting.*

Answer:



3 Gate Teleportation in ZX

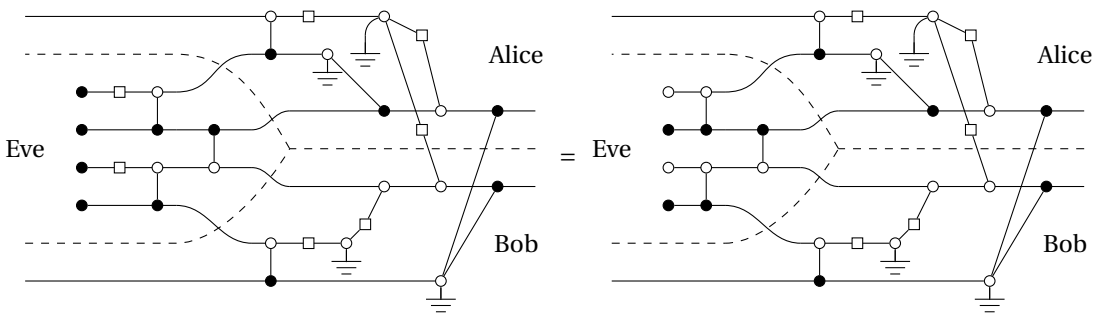
Alice and Bob both possess a qubit, and they would like to apply a CNot gate between the two, but they can only communicate classical information. The problem can be solved if they initially share an entangled state (created by Eve in the following circuit). This entangled state, and the rest of the protocol are represented below:

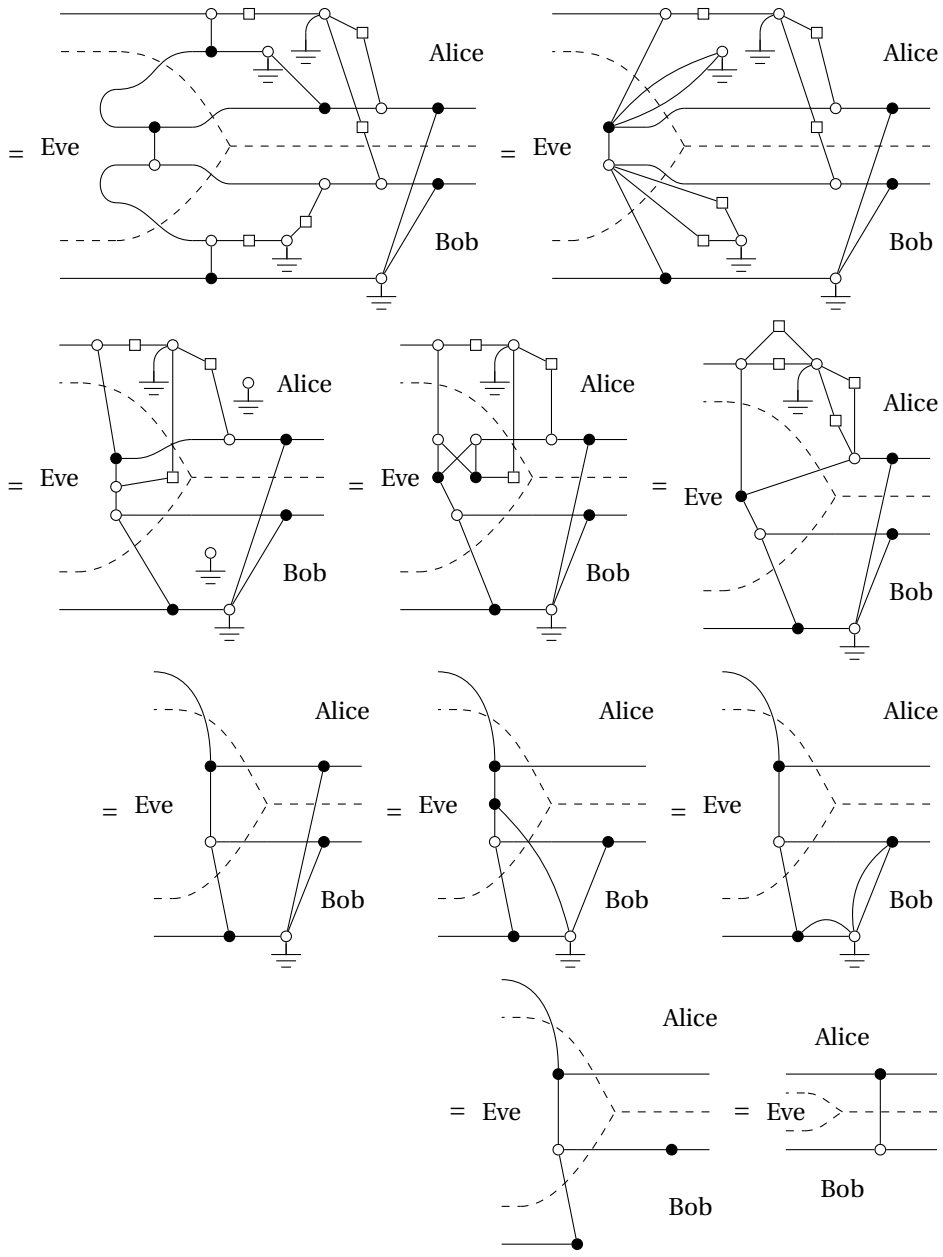


Question 1. Turn the above protocol into a ZX-diagram with discards (not variables).

Question 2. By rewriting the diagram, show that Eve has applied a CNot gate between Alice's qubit and Bob's qubit. *Hint: You can reuse the result of Question 1 of Exercise 1, even if you have not proved it. Hint: You can try and simplify using axioms or derived equations from the course. If your are stuck, using the Clifford normal form (with discards treated as outputs) should help.*

Answer:





4 A CSS state

Question 1. How many stabilisers do you need to represent a CSS state on 6 qubits?

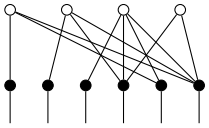
Answer: We need 6 stabilisers total, but for a given set of $n \leq 6$ X-stabilisers, if those are all the X-stabilisers, the $6-n$ Z-stabilisers are induced.

Consider the following X parity check matrix:

$$H_X = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

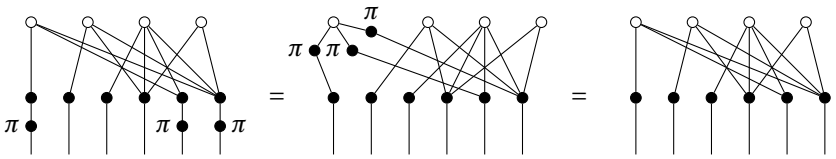
Question 2. Build the 6-qubit ZX-state associated with the above X parity check matrix.

Answer:



Question 3. Recall how to represent the X gate in ZX. Graphically show that XIIIXX indeed stabilises the state.

Answer:



Question 4. What should be the dimensions of an associated Z parity check matrix H_Z for the H_X above? Give one such H_Z.
Hint: You can try reasoning on the matrices themselves, or on the ZX-diagram.

Answer: Since we have 4 X-stabilisers, we need $6-4$ Z-stabilisers. The dimensions of H_Z should hence be 2×6. One possible such H_Z is:

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

5 Stabiliser state

Question 1. Consider the group generated by the following Pauli strings: $-XYXI$, $YXIY$, $-IZYX$, $-YYZX$. Briefly explain whether $-IIII$ is part of the group. Give the associated parity check matrix.

Answer: To check whether $-IIII$ is part of the group or not, it suffices to check whether the stabilisers all commute (and that they are all involutive, i.e. their scalar factor is 1 or -1). This is indeed the case: the number of overlapping, different, non-I Paulis in each pair of stabilisers is even. Taking each stabiliser in the given order, the associated parity-check matrix is:

$$\left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{array}\right)$$

Question 2. Simplify the parity check matrix into an equivalent one, and out of the result, build a ZX-state stabilised by the given Pauli strings.

Answer:

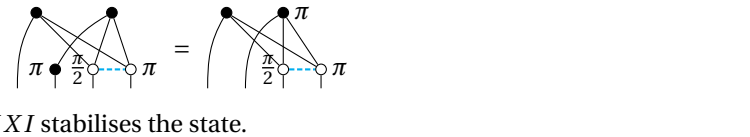
$$\left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \end{array}\right) \equiv \left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right) \equiv \left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 3 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right)$$

$$\equiv \left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right) \equiv \left(\begin{array}{c|cccc|cccc} 3 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right) \equiv \left(\begin{array}{c|cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 2 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right)$$

The ZX-state can then be computed as follows:

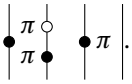
$$\left(\left(\begin{array}{c|cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 2 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right), \left|\quad\right|\quad\left|\quad\right|\quad\left|\quad\right|\right) \equiv \left(\left(\begin{array}{c|cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right), \left|\quad\right|\quad\left|\quad\right|\quad\right) \equiv \left(\left(\begin{array}{c|cccc|cccc} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right), \left|\quad\right|\quad\left|\quad\right|\right)$$

$$\equiv \left(\left(\begin{array}{c|c|c} 0 & 1\,1\,1\,0 & 0\,0\,0\,0 \\ 0 & 1\,1\,0\,1 & 0\,0\,0\,0 \\ \hline 0 & 0\,0\,0\,0 & 1\,0\,1\,1 \\ 2 & 0\,0\,0\,0 & 0\,1\,1\,1 \end{array} \right), \left| \quad \right| \right) \equiv$$



at which point the remaining stabilisers are CSS stabilisers. Building the corresponding CSS state on top of the obtained diagram yields:

Question 3. Graphically show that $-XYXI$ stabilises the state.

Answer: Since $Y = iXZ$, we can represent the stabiliser above as $-i \cdot$  . Then:

